

Lambda or ECS for document processing?

Assurance Horizon · Insurance · Level 2

36 employees · 3 engineers Current step: portfolio 90/100 · needs_revision

 This is a synthetic consulting case created for cloud architecture practice. No real customer data is represented.

01 / DISCOVERY

Discovery

Business brief

An insurance broker wants customers to upload documents and receive an extraction result without waiting on a web request.

What the client initially shared

- The client needs a justified cloud architecture decision and a feasible transition plan.
- Interview the client to establish workload, service targets, budget and operational constraints.

Stakeholders

Alex Morgan
Business sponsor

Customer impact and business priorities

Priya Shah
CTO

Feasibility, reliability, and engineering capacity

Sam Chen
Lead engineer

How the current system behaves and who operates it

Jordan Ellis
CFO

reversibility

Skip to content

Questions and answers

My question

How many documents are processed per day, what is the peak upload rate, and how long does one document usually take to process?

Alex Morgan

I don't have a reliable answer to that yet. Could you narrow the question to a business constraint or a specific measurement? If we aren't measuring it, we can discuss what evidence you need.

My question

What is the maximum acceptable time between a customer uploading a document and receiving the extraction result?

Alex Morgan

I don't have a reliable answer to that yet. Could you narrow the question to a business constraint or a specific measurement? If we aren't measuring it, we can discuss what evidence you need.

My question

What is the largest document size, in MB, that the system must support?

Alex Morgan

PDF processing takes 30 seconds normally, but scanned bundles take 24 minutes; uploads arrive in bursts. (Source: Lead engineer)

My question

What percentage of documents are scanned bundles that take around 24 minutes to process?

Alex Morgan

I don't have a reliable answer to that yet. Could you narrow the question to a business constraint or a specific measurement? If we are an discuss what evidence you need.

[Skip to content](#)

My question

Must a 24-minute scanned bundle be processed as one continuous job, or can it be split into smaller steps?

Alex Morgan

I don't have a reliable answer to that yet. Could you narrow the question to a business constraint or a specific measurement? If we aren't measuring it, we can discuss what evidence you need.

My question

Provide logs for the longest document-processing jobs, including processing time, failures, and document type.

Alex Morgan

I don't have a reliable answer to that yet. Could you narrow the question to a business constraint or a specific measurement? If we aren't measuring it, we can discuss what evidence you need.

My question

What happens when a document fails to process? Should the system retry automatically, and how many retries are acceptable?

Alex Morgan

PDF processing takes 30 seconds normally, but scanned bundles take 24 minutes; uploads arrive in bursts. (Source: Lead engineer)

The selected standard Lambda execution model has a 15-minute invocation limit, below the longest bundle duration. The median alone hides this constraint; changing execution models requires a separate compatibility review. (Source: Lead engineer)

My question

What is the highest number of documents uploaded within a 5-minute period?

Alex Morgan

I don't have a reliable answer to that yet. Could you narrow the question to a business constraint or a specific measurement? If we are

[Skip to content](#)

My question

Is it acceptable for a document to wait up to 5 minutes in a queue before processing starts?

Alex Morgan

PDF processing takes 30 seconds normally, but scanned bundles take 24 minutes; uploads arrive in bursts. (Source: Lead engineer)

The selected standard Lambda execution model has a 15-minute invocation limit, below the longest bundle duration. The median alone hides this constraint; changing execution models requires a separate compatibility review. (Source: Lead engineer)

My question

What is the maximum monthly cloud budget for document processing?

Alex Morgan

Results within an hour; durable uploads; peak 400 bundles/hour; one operations engineer. These are planning constraints; compare actual estimates including operations, data transfer and resilience. No supplier quote has been approved. (Source: Budget owner)

My question

How many hours per week can the operations engineer spend managing this document-processing system?

Alex Morgan

Jobs are restartable only at document boundaries. A progress record exists but retries currently create duplicate outputs. (Source: Operations owner)

3 engineers maintain the product. Confirm who owns deployment, patching, incidents and recovery before committing to an option. Results within an hour; durable uploads; peak 400 bundles/hour; one operations engineer. (Source: Engineering manager)

My question

Who owns deployments, patching, incidents, and recovery, and how many hours per week can they spend operating this system?

[Skip to content](#)

Alex Morgan

PDF processing takes 30 seconds normally, but scanned bundles take 24 minutes; uploads arrive in bursts. (Source: Lead engineer)

Results within an hour; durable uploads; peak 400 bundles/hour; one operations engineer. These are planning constraints; compare actual estimates including operations, data transfer and resilience. No supplier quote has been approved. (Source: Budget owner)

My question

Provide the last 30 days of incident, deployment, and maintenance records for the document-processing system.

Alex Morgan

PDF processing takes 30 seconds normally, but scanned bundles take 24 minutes; uploads arrive in bursts. (Source: Lead engineer)

Jobs are restartable only at document boundaries. A progress record exists but retries currently create duplicate outputs. (Source: Operations owner)

My question

Must the system prevent duplicate extraction results when a job is retried?

Alex Morgan

PDF processing takes 30 seconds normally, but scanned bundles take 24 minutes; uploads arrive in bursts. (Source: Lead engineer)

Evidence requests

No evidence requests recorded yet.

Facts, assumptions, and unknowns

Type	Note	Source / confidence	Validation
Fact	PDF processing takes 30 seconds normally, but scanned bundles take 24 minutes; uploads arrive in bursts	Lead engineer · medium	—
Unknown	We do not know if the 24-minute scanned bundles can be split into Skip to content assing jobs.	chief · high	Review the processing workflow with the engineering team and test whether long scanned bundles can be split into smaller steps.

Type	Note	Source / confidence	Validation
Fact	The standard Lambda model has a 15-minute execution limit, which is shorter than the longest processing time.	Lead engineer · high	—
Unknown	We do not know the peak upload volume during bursts.	chief · high	Review upload logs from the last 30 days and measure the highest 5-minute upload volume.
Fact	Results within an hour; durable uploads; peak 400 bundles/hour; one operations engineer. These are planning constraints; compare actual estimates including operations, data transfer and resilience. No supplier quote has been approved.	Budget owner · high	—
Fact	Jobs can only restart at document boundaries.	Lead engineer · high	—
Fact	A progress record already exists.	Lead engineer · high	—
Fact	Retries currently create duplicate outputs.	Lead engineer · high	—
Fact	Three engineers maintain the product, including one operations engineer.	Lead engineer · high	—

02 / PROBLEM FRAMING

Problem framing

Stated Problem	An insurance broker wants customers to upload documents and receive an extraction result without waiting on a web request.
Actual Problem	The client needs an asynchronous document-processing system that can handle short and long jobs, traffic bursts, and retries without blocking the customer.
Business Impact	A poor design could delay customer results, lose uploaded documents, create duplicate outputs, or increase operating cost. The system must return results within one hour.
 Skip to content	Normal PDFs take about 30 seconds, while scanned bundles can take 24 minutes. Uploads arrive in bursts of up to 400 bundles per hour. Standard Lambda has a 15-minute execution limit. Retries

currently create duplicate outputs.

Constraints	Results must be available within one hour. Uploads must be durable. The system must support peaks of 400 bundles per hour. Three engineers maintain the product, including one operations engineer. Costs must include operations, data transfer, and resilience.
Uncertainties	We do not know if the 24-minute jobs can be split into smaller steps, the real operational effort required, or the final monthly cost of each option. These points could change the architecture decision.

03 / REQUIREMENTS

Business and technical requirements

ID	Requirement	Priority	Source
R01	The solution must support document processing jobs that can last up to 24 minutes.	must	Lead engineer
R02	Results must be available within 1 hour. Uploads must be durable and not lost. The system must support peaks of 400 bundles per hour. Cost estimates must include operations, data transfer, and resilience.	must	Budget owner

04 / ARCHITECTURE OPTIONS

Architecture options

OPTION 1	
Lambda for document processing	
Summary	Store uploaded documents durably, place processing requests in a queue, and use Lambda functions to process documents asynchronously.
Components	Documents are stored in durable object storage. A queue receives processing jobs. Lambda functions process documents asynchronously.
Skip to content	Results and progress are stored separately. Failed jobs can be retried.

Advantages	Easy to scale during upload bursts. No servers to manage. Good fit for short 30-second PDF jobs. Low operational burden for the small team.
Disadvantages	The standard Lambda execution limit is 15 minutes, while some scanned bundles take 24 minutes. Long jobs would need to be split or handled differently. Retry logic must prevent duplicate outputs.
Cost	Cost can be efficient for short and bursty workloads because compute is used only when processing runs. Long-running jobs may require another execution model.
Operations	Low to medium. AWS manages the compute, but the team still needs queueing, retry, progress, and duplicate-output handling.
Reliability	Durable storage and queueing can protect uploads and allow failed jobs to retry. The design must make retries safe.
Security	Documents and results need strong access control and encryption. Functions should only have the permissions they need.
Implementation	Medium. Short jobs are simple, but 24-minute bundles and duplicate-safe retries add complexity.
Assumptions	Most jobs are short enough for Lambda. Long bundles can be split or moved to another processing model.
Risks	A 24-minute bundle cannot run as one standard Lambda invocation. Retries may create duplicate results if idempotency is not added.
Tradeoff Rationale	Lambda fits the bursty workload and reduces operational work for the small team. It is also a good fit for the normal 30-second jobs. However, the 15-minute execution limit cannot support the 24-minute scanned bundles as one job, so the long-running requirement is at risk.

[Skip to content](#)

ECS for document processing

Summary	Store documents durably, place jobs in a queue, and use ECS workers to process both short PDFs and long scanned bundles asynchronously.
Components	Documents are stored in durable object storage. A queue holds processing jobs. ECS workers read jobs from the queue and process documents. Workers can scale when upload volume increases. Progress and final results are stored separately.
Advantages	Supports both 30-second jobs and 24-minute jobs. No 15-minute processing limit. More control over CPU, memory, libraries, and runtime. One processing model can handle all document types.
Disadvantages	More infrastructure to operate than Lambda. Capacity and scaling must be managed carefully. The small operations team may have more work.
Cost	ECS may cost more during low usage because some compute capacity may remain running. It may be more predictable for long-running workloads.
Operations	Medium to high. The team must manage container capacity, scaling, deployments, monitoring, and recovery.
Reliability	Queueing allows jobs to survive worker failures. Workers can restart processing, but duplicate-output handling is still required.
Security	The team must secure container tasks, permissions, networking, images, and stored documents.
Implementation	Medium. It supports the long jobs naturally, but requires more infrastructure and operational setup.
Assumptions	The team can operate ECS with the available engineering capacity. The workload justifies the additional infrastructure.
Skip to content	Operational work may be too high for one operations engineer. Poor scaling could increase cost or delay processing during the 400-bundle-per-hour peak.

Tradeoff Rationale	ECS can process both 30-second PDFs and 24-minute scanned bundles without splitting the jobs. It also gives more control over compute resources. However, it creates more operational work for a team with only one operations engineer and may cost more when capacity is underused.
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Trade-off matrix

Dimension	Lambda for document processing	ECS for document processing
Cost	4/5	3/5
Reliability	2/5	4/5
Performance	3/5	4/5
Scalability	5/5	4/5
Security	4/5	3/5
Operational Burden	5/5	2/5
Implementation Speed	3/5	4/5
Team Fit	5/5	2/5
Flexibility	2/5	5/5
Reversibility	4/5	4/5

Contextual ratings by the learner · 1–5, higher is more favorable · not measured benchmarks.

Requirement traceability

Requirement	Decision	Status	Rationale
R01	Use ECS workers for document processing so jobs can run longer than 15 minutes. Status of R01	Satisfied	ECS can process the 24-minute scanned bundles without splitting them into smaller jobs.
R02	Store uploads durably, place jobs in a queue, and use ECS workers during bursts. Compare full cost including operations, data transfer, and Skip to content	Unaddressed	The design supports durable uploads, burst handling, and the one-hour result target, but the real monthly cost and operational effort have not been validated yet.

What would change my mind?

01	Condition	The 24-minute scanned bundles can be safely split into smaller jobs that work with Lambda.
	Signal	Processing time, failure rate, duplicate outputs, and total cost during a Lambda test.
	Threshold	All processing steps finish within the Lambda limit, meet the 1-hour result target, and do not create duplicate outputs.
	Alternative	Move document processing from ECS to Lambda.

Validation and proof of concept

Required	Yes
Justification	Not recorded.
Hypothesis	ECS can process both short PDFs and 24-minute scanned bundles, handle burst traffic, and return results within one hour without losing uploads or creating duplicate outputs.
Scope	Test durable uploads, the processing queue, ECS worker scaling, 24-minute jobs, retries, idempotency, and repeated failures.
Metrics	Processing time, queue delay, worker scaling time, failed jobs, duplicate outputs, retry success rate, and estimated monthly cost.
Success	All uploads are preserved, 24-minute jobs complete successfully, results are returned within one hour, burst traffic is processed without major delay, and retries do not create duplicate outputs.
Exit Skip to content	Stop if uploads are lost, 24-minute jobs fail, results exceed the one-hour target, retries create duplicates, or the operational cost is too high.

Go No Go

Go if ECS meets the time, durability, burst, retry, and cost requirements. No-Go if one of these key requirements cannot be met.

ADR working draft

Title	ECS for asynchronous document processing
Context	Normal PDFs take about 30 seconds, while scanned bundles can take up to 24 minutes. Uploads arrive in bursts of up to 400 bundles per hour. Results must be available within one hour, uploads must be durable, and retries currently create duplicate outputs.
Decision	Use durable storage and a processing queue, then run document-processing workers on ECS. Scale workers during bursts and use idempotency to prevent duplicate outputs.
Alternatives	Lambda was considered because it is simpler to operate and scales easily. It was not selected because the standard Lambda model has a 15-minute execution limit, while some jobs take 24 minutes.
Consequences	ECS supports both short and long jobs and gives more control over processing. The trade-off is more operational work for scaling, deployments, monitoring, and recovery.
Risks	The main risks are higher operational effort for the small team, higher cost if workers are underused, and processing delays if scaling is too slow during large bursts.

Accepted decision records

ADR-001 · ECS for asynchronous document processing

Context

Normal PDFs take about 30 seconds, while scanned bundles can take up to 24 minutes. Uploads arrive in bursts of up to 400 bundles per hour. Results must be available within one hour, uploads must be durable, and retries currently create duplicate outputs.

Decision

Use durable storage and a processing queue, then run document-processing workers on ECS. Scale workers during bursts and use idempotency to prevent duplicate outputs.

Skip to content

Alternatives	Lambda was considered because it is simpler to operate and scales easily. It was not selected because the standard Lambda model has a 15-minute execution limit, while some jobs take 24 minutes.
Consequences	ECS supports both short and long jobs and gives more control over processing. The trade-off is more operational work for scaling, deployments, monitoring, and recovery.
Risks	The main risks are higher operational effort for the small team, higher cost if workers are underused, and processing delays if scaling is too slow during large bursts.
Requirements	R01
Reason	Not recorded.
Status	accepted

11 / COMMUNICATION

Communicating the decision

Engineer	Documents are stored first, then placed in a queue. ECS workers process the jobs and scale during bursts. We also use idempotency to prevent duplicate outputs and a dead-letter queue for repeated failures.
Cto	The design supports both 30-second PDFs and 24-minute scanned bundles. It also separates uploads from processing, which improves resilience. The main trade-off is more operational work than Lambda.
Cfo	ECS can handle all job durations, but we must validate the full monthly cost, including idle capacity, data transfer, resilience, and engineering effort.
Ceo	Customers can upload documents without waiting for processing to finish. The system is designed to keep uploads safe, handle traffic peaks, and return results within one hour.
Analogy	Think of the queue like a waiting line in a processing center. Documents enter the line, and workers take them one by one. When more documents arrive, more workers can be added.
Skip to content	

Customer uploads are stored safely first. Processing requests are then placed in a queue, and a group of workers processes them in the background. More workers can be added during busy periods, failed jobs can be retried, and duplicate results are prevented.

Review and responses

Reviewer: Mock · local simulation

Decision-critical evidence · critical

R02 is a must-have requirement with no addressed decision. Record a mitigation or revise the recommendation.

Response: accept

Rationale: I agree. R02 is a must-have requirement. Before production, we need to validate the full monthly cost, operational effort, data transfer, and resilience. If the cost or operational effort is too high, we should review the recommendation.

Facts versus assumptions · warning

Your notes do not distinguish both assumptions and unknowns. Identify what is inferred, what remains unmeasured, and which uncertainty could change this decision.

Response: accept

Rationale: I agree. The 24-minute jobs and 400 bundles per hour are facts. The real monthly cost and operational effort are still unknown and could change the decision.

[Skip to content](#)

Validation · warning

No evidence request supports the recommendation yet. Defend why the available client statements are enough, or identify one measurement that could disprove your diagnosis.

Response: accept

Rationale: I agree. We still need evidence. A full load test at 400 bundles per hour and a full-month cost estimate could prove the recommendation wrong.

Trade-offs · info

You recommend “Use durable storage and a queue for uploads, then use ECS workers to process documents asynchronously. Scale the workers during bursts and store progress and results separately.”. Defend it against your least complex alternative using a specific requirement and the team’s operating capacity. A more sophisticated service is not automatically a better fit.

Response: partial

Rationale: I partially agree. Lambda is simpler for the small team, but the selected Lambda model cannot run the 24-minute jobs. This is why ECS is currently preferred.

Change of mind · info

Your first reconsideration threshold is “All processing steps finish within the Lambda limit, meet the 1-hour result target, and do not create duplicate outputs.”. Explain who will observe this signal and when the decision will be reviewed.

Response: accept

Rationale: I agree. The engineering team should test the Lambda alternative before production. If it handles the long jobs, meets the one-hour target, and avoids duplicates, we should review the decision.

[Skip to content](#)

Communication · CEO / CFO · info

Your executive explanation includes service acronyms. Translate them into financial exposure, customer outcomes, and the trade-off the executive must approve.

Response: accept

Rationale: I agree. The CEO and CFO need to understand customer impact, total cost, and operational risk instead of cloud service names.

13 / FINAL REVISION

Final decision and lessons

Final Decision

I keep the ECS recommendation because it supports the confirmed 24-minute jobs and the peak workload. After review, I would not approve production until we validate the full monthly cost and operational effort. If Lambda can support the long jobs safely or ECS becomes too expensive to operate, I would review the decision.

Lessons

I learned to connect each client requirement directly to the architecture design. This makes the decision easier to explain because I can show the client why each part of the architecture exists and which business need it supports.

EVALUATION

Evaluation

90/100

Needs revision

Transparent rubric · scenario rules

[Skip to content](#)

Critical misses

- R02 is a must-have requirement with no addressed decision. Record a mitigation or revise the recommendation.

Dimension	Score	Feedback
Discovery quality	13/20	5/6 discovery topics established; 0 reasoned evidence request(s). Keep assumptions separate from facts and pursue constraints that could change the decision.
Problem framing	10/10	A strong framing separates the client’s symptoms from an actionable business problem and makes uncertainties explicit.
Architecture	15/15	2 alternatives compared; 100% of requirements traced. The rubric rewards documented fit, not particular cloud services.
Trade-off reasoning	20/20	Trade-offs need explicit advantages, accepted costs, simpler alternatives, and connections to requirements. Numeric ratings do not determine this score.
Cost and business reasoning	7/10	Tie costs to a stated budget, observed baseline, or financial impact. Ranges and explicitly labeled estimates are acceptable.
Security, reliability, operations	10/10	Document failure modes, security controls, operational ownership, and a practical validation or exit path.
Communication	10/10	Adapt the same decision to each audience. Executive explanations should describe business outcomes and risk in plain language.
Limits and change of mind	5/5	Revisit the decision when an observable threshold changes. A POC should have measurable success and stop criteria.

REASONING HISTORY

How the work changed

2 immutable snapshot(s) are saved.

First analysis

Final revision

Field	Before review	Final revision
Final Decision	Not recorded	I keep the ECS recommendation because it supports the confirmed 24-minute jobs and the peak workload. After review, I would not approve production until we validate the full monthly cost and operational effort. If Lambda can support the long jobs safely or ECS becomes too expensive to operate, I would review the decision.
⌕ Skip to content		I learned to connect each client requirement directly to the architecture design. This makes the decision easier to explain because I can show the client why each part of the architecture exists and which business need it supports.

Publication status

Published in the portfolio.

This PDF reflects the saved case.